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MILITARY SPECIFICATION

CARD, TANK RANGE: 8724207

1. SCOPE

1.1 This specification covers one type of range card for recording pertinent target data for tanks.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926 —Fire Control Materiel: General Specification Governing the Manufacture and Inspection of.
MIL-P-14232 —Parts, Equipment and Tools for Ordnance Materiel; Packaging of.
MIL-I-45208 —Inspection Requirements, General Specification for.

STANDARDS

MILITARY

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109—Quality Assurance Terms and Definitions.

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

D8724207 —Card, Tank Range: 8724207.
F8598890 —Test Fixture, Luminosity.
IEL 8724207—Index of Inspection Equipment Numbers for Card, Tank Range: 8724207.

PACKAGING DATA SHEET

MIL-P-14232/P8724207—Packaging of Card, Tank, Range: 8724207

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications forming a part of this specification.

3.2 Fabrication. The range card shall be manufactured in accordance with Drawing D8724207 and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to insure inherent quality and performance capabilities.

3.3 General specifications. The range card shall meet the following requirements of Specification MIL-F-13926.

FSC 1220

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(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Part identification and marking.
(e) Workmanship.

3.4 Environmental.

3.4.1 *Storage temperature.* When placed in a flat condition the range card shall be subjected to ambient temperatures of plus 140 and minus 80 degrees Fahrenheit until thermal stabilization is reached and upon return to standard ambient temperature (60 to 90 degrees Fahrenheit) it shall show no evidence of cracks, separation of laminations or sheets, warpage from the original flat condition, or permanent deformation beyond $\frac{3}{32}$ inch and shall subsequently meet all the following requirements of this specification at the standard ambient temperature.

3.4.2 *Shock resistance.* With the range card held vertically and the lower edge horizontally, it shall be dropped onto a hard surface from a height of three feet and shall show no evidence of chipping, separation of laminations or sheets, shattering or cracking as a result thereof. The range card shall also meet the above requirements when held in a horizontal position.

3.5 Performance.

3.5.1 *Luminescence.* The area of the range card specified as being phosphorescent (front and back) in accordance with Drawing D8724207 shall be opaquely masked except for a circular area one inch in diameter, in a clear location, which shall be exposed for a period of sixty seconds to an incandescent tungsten lamp positioned to deliver an illumination equal to five hundred foot candles on the unmasked surface. After irradiation the minimum luminosity values shall conform to Table I.

TABLE I. *Luminosity*

Elapsed time after irradiation in minutes	*Luminosity in effective microlamberts*
2	80.0
5	30.0
10	12.5
30	3.3
60	1.2
120	0.5

3.5.2 *Legibility.* Both sides of the range card shall have matte surfaces capable of accepting clear and legible markings from a lead pencil of a hardness of 2H, and upon erasure with a commercial pencil eraser, there shall be no evidence of permanent marks, smudges or smears.

3.5.3 *Markings.* Diagram lines and lettering shall be permanent, black in color, and clearly legible in emitted and reflected light.

3.5.4 *Deflection resistance.* The range card shall be deflected a minimum of three times each side, alternately front and back, with the center of the card supported by a one inch diameter rod and two corners simultaneously deflected at least one and one-quarter inches below the center contact point of the rod. Subsequent to the deflection there shall be no cracks, separation or permanent deformation of laminations or sheets or other noticeable defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *General provisions.* The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL–I–45208 and the component inspection requirements of Specification MIL–F–13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL–STD–109.

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4.2 Inspection provisions.

4.2.1 *Inspection lot.* Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" as specified in Standard MIL–STD–105.

4.2.2 *Examination and tests.* Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL–F–13926. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL–STD–105, and the inspection level and sampling plans specified in Tables II and III titled "Classification of Defects". Examinations and tests for Packaging, Packing, and Marking shall be in accordance with Specification MIL–P–14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual nonconforming items.

TABLE II. *Classification of Defects*

Use Level S–2 of Table I with Sampling Plan Table II–A of Standard MIL–STD–105.

Critical: None.

Major: AQL 2.5% defective.

	Requirement	*Test procedure*
102. Luminescence	3.5.1	4.5.3
102. Deflection resistance	3.5.4	4.5.6

TABLE III. *Classification of Defects*

Use Level II of Table I with Sampling Plan Table II–A of Standard MIL–STD–105.

Critical: None.

Major: AQL 2.5% defective.

	Requirement	*Test procedure*
101. Legibility	3.5.2	4.5.4

Minor: AQL 4.0% defective.

201. Markings	3.5.3	4.5.5

4.2.3 *Disposition of non-conforming product.* Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in Standard MIL–STD–105.

4.3 Control tests.

4.3.1 *General.* One range card shall be selected at random from each 300 produced, or from each months production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

Control test	*Requirement*	*Test procedure*
(a) Materials	3.1	Visual—Drawings, Specifications
(b) Fabrication	3.2	Visual—Drawing
(c) General specification	3.3	Visual—Standard Measuring Equipment—MIL–F–18926

4.3.2 *Environmental and functional.* Three range cards shall be selected at random from each 100 produced, or from each months production, whichever occurs first. Control samples have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

Control test	*Requirement*	*Test procedure*
(e) Storage temperature	3.4.1	4.5.1
(f) Shock resistance	3.4.2	4.5.2
(g) Luminescence	3.5.1	4.5.3

4.3.3 *Control test failure.* Should the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Inspection Equipment" requirement of Specification MIL–I–45208. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 *Government furnished inspection*

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equipment. Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with "Inspection Equipment", and "Government Furnished Material" requirements of Specification MIL–I–45208.

4.4.2 *Contractor furnished inspection equipment.*

4.4.2.1 *Government design.* All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the List of Inspection Equipment Numbers, IEL–8724207. Alternate designs may be substituted only as provided by "Inspection Provisions" of Specification MIL–I–45208.

4.4.2.2 *Contractor design.* The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL–F–13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.2.3 *Equipment to test luminescense requirement.* Equipment manufactured in accordance with the design depicted on Drawing F8598890 Test Fixture, Luminosity, shall be mandatory for Government acceptance tests of the Tank Range Card 8724207.

4.4.2.4 *Equipment to test temperature requirement.* The environmental testing equipment used to determine conformance with the temperature requirement shall conform to the "Test Facilities" requirements contained in Specification MIL–F–13926.

4.5 Test methods and procedures.

4.5.1 *Storage temperature.* The range card shall be exposed to the temperature indicated in 3.4.1 utilizing test facilities conforming with the requirements of 4.4.2.4. This equipment shall be capable of maintaining the specified temperatures until thermal stabilization is reached. The temperatures shall be changed gradually to avoid thermal shock. After the card has been exposed to the extreme temperatures and permitted to return to standard ambient temperature, the card shall be visually inspected for damage indicated in 3.4.1. Any warpage of the card shall be measured when the card is placed in a flat position on a surface plate. Measurements shall be made using standard measuring equipment. Measurements noted shall not exceed the tolerance specified. The card shall then be subjected to the tests of 4.5.2 to 4.5.6 inclusive.

4.5.2 *Shock resistance.* When held in the vertical position, the range card shall be subjected to one free fall on to a hard surface (steel plate at least ½ inch thick or a concrete slab) to be followed by one free fall when held in the horizontal position. The two falls shall be free-falls from the specified height. The drop height shall be measured from the lowermost edge of the card to the hard surface. Immediately following the shock test the card shall be visually inspected for damage and shall meet the requirement specified in 3.4.2.

4.5.3 *Luminescence.* Luminosity values of the Range Card shall be measured on the special testing device specified in 4.4.2.3. This test shall be conducted in accordance with the set-up and operating instructions outlined in Drawing F8598890. The control sample shall be tested for all the requirements of Table I. The samples selected from production (Table II) shall be tested only at 2 minutes and 120 minutes after irradiation. The range card shall meet the minimum required luminosity values specified in 3.5.1.

4.5.4 *Legibility.* The surface of the range card shall be capable of accepting clear and legible markings from a lead pencil of a hard-

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ness 2H under normal writing pressure and upon erasure with a commercial pencil eraser there shall be no evidence of the defects specified in 3.5.2. The pencil markings shall be applied to the four quadrants, each side, of the range card.

4.5.5 *Markings.* The lines and lettering of the range card shall meet the requirement specified when visually examined in an illuminated area, followed by a visual examination in a blacked-out area.

4.5.6 *Deflection resistance.* This test shall be conducted with the aid of a one inch diameter rod, at least two inches in length. This rod shall be positioned, on end, on a flat surface. The range card shall be positioned so that its center is supported by the rod and the two opposite corners of the card deflected simultaneously as specified. Subsequent to these deflections, the range card shall show no evidence of the defects specified in 3.5.4.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Specification MIL–P–14232 and Packaging Data Sheet MIL–P–14232/P8724207. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The tank range card is utilized to record thereon in pencil, target positions, terrain data and other pertinent information as observed in daylight hours for reference to aid in redetermining those positions after dark.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title number and date of this specification

(b) Selection of applicable levels of preservation, packaging and marking (5.1)

(c) Applicable packaging Data Sheet Numbers (see 5.1)

(d) Applicable stock number

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